

CN - Lab03 Tasks Submission

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Task 01:

Server Code:

```
import socket
import json
s= socket.socket()
s.bind(('localhost', 9999))
s.listen(3)
print("Waiting for connections")

c,address = s.accept()

data = c.recv(1024).decode()

message = json.loads(data)

num1 = message["num1"]
num2 = message["num2"]
operation = message["operation"]

if operation == "+":
    result = num1+num2

elif operation == "-":
    result = num1-num2
elif operation == "*":
    result = num1*num2
elif operation == "/":
    if num2 ==0:
        result = "Error: Division by zero"
    else:
        result = num1/num2
```

```
else:
    result = "Error: Invalid operation"

details = {
    "num1": num1,
    "num2": num2,
    "operation": operation,
    "result": result
}

with open("Lab03Task01Server.json", "w") as f:
    json.dump(details, f, indent=4)

response = str(result)
c.send(response.encode())
c.close()
s.close()
```

Output:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\yashm\Desktop\CN LAB03> python -u "c:\Users\yashm\Desktop\CN LAB03\Lab03Task01Server.
Waiting for connections
● PS C:\Users\yashm\Desktop\CN LAB03> python -u "c:\Users\yashm\Desktop\CN LAB03\Lab03Task01Server.
Waiting for connections
Got connection from ('127.0.0.1', 54811)
❖ PS C:\Users\yashm\Desktop\CN LAB03> 
```

Client Code:

```
import socket
import json

s= socket.socket(socket.AF_INET,socket.SOCK_STREAM)
s.connect(('localhost', 9999))

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
operation = input("Enter operation (+, -, *, /): ")

data = {
    "num1": num1,
    "num2": num2,
    "operation": operation
}

message = json.dumps(data)
s.send(message.encode())

result = s.recv(1024).decode()
print("Result:", result)
```

Output:

```
PS C:\Users\yashm\Desktop\CN LAB03> python Lab03Task01Client.py
Enter first number: 3
Enter second number: 9
Enter operation (+, -, *, /): *
Result: 27
❖ PS C:\Users\yashm\Desktop\CN LAB03> 
```

Task 02:

Server Code:

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind(('localhost', 9999))
print("Socket successfully created and bound to port 9999")
s.listen(5)
print("Server is listening on port 9999...")

c, address = s.accept()
print("Got connection from", address)

message = c.recv(1024).decode()

grade_point = float(message)
if grade_point == 4.33:
    grade = "A+"
    qualification = "Excellent"

elif grade_point == 4.00:
    grade = "A"
    qualification = "Excellent"

elif grade_point == 3.66:
    grade = "A-"
    qualification = "Very good"

elif grade_point == 3.33:
    grade = "B+"
    qualification = "Very good"

elif grade_point == 3.00:
    grade = "B"
    qualification = "Very good"

elif grade_point == 2.66:
    grade = "B-"
    qualification = "Good"
```

```

elif grade_point == 2.33:
    grade = "C+"
    qualification = "Good"

elif grade_point == 2.00:
    grade = "C"
    qualification = "Good"

elif grade_point == 1.66:
    grade = "C-"
    qualification = "Passable"

elif grade_point == 1.33:
    grade = "D+"
    qualification = "Passable"

elif grade_point == 1.00:
    grade = "D"
    qualification = "Passable"

elif grade_point == 0.00:
    grade = "E"
    qualification = "Failure"
else:
    grade = "Invalid"
    qualification = "Invalid grade point"

response = f"Grade: {grade}, Qualification: {qualification}"
c.send(response.encode())
c.close()
s.close()

```

Output:

```

PS C:\Users\yashm\Desktop\CN LAB03> python -u "c:\Users\yashm\Desktop\CN LAB03\Lab03Task02
Socket successfully created and bound to port 9999
Server is listening on port 9999...
Got connection from ('127.0.0.1', 59752)
PS C:\Users\yashm\Desktop\CN LAB03>

```

Client Code:

```
import socket

s= socket.socket()
print("Socket successfully created")

Points = float(input("Enter your points: "))
s.connect(('localhost',9999))
data = {
    "Points": Points}
s.send(str(Points).encode())

response = s.recv(1024).decode()
print("Response from server:", response)
s.close()
```

Output:

```
PS C:\Users\yashm\Desktop\CN LAB03> python Lab03Task02Client.py
Socket successfully created
Enter your points: 3.33
Response from server: Grade: B+, Qualification: Very good
PS C:\Users\yashm\Desktop\CN LAB03>
```